Positive-Velocity Goldfish Flow: Global Root Interlacing and One-Carrier Scattering

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Abstract

For arbitrary ordered real positions and strictly positive velocities, we prove that the Calogero goldfish polynomial pencil has simple real roots at every real time. A decreasing partial-fraction map gives exact signed-time interlacing, global collision avoidance, and strict coordinate monotonicity.

1 Polynomial linearization and global interlacing

Fix $N \geq 2$, positions $x_1 < \cdots < x_N$, and velocities $v_i > 0$. Define

$$P(z) = \prod_{i=1}^{N}(z - x_i), \qquad Q(z) = \sum_{i=1}^{N} v_i \prod_{j\neq i}(z - x_j), \qquad p(z,t) = P(z) - tQ(z). \tag{1}$$

Let $y_1 < \cdots < y_{N-1}$ denote the roots of Q.

Theorem 1 (Complete positive-cone flow)**.** *For every $t \in \mathbb{R}$, $p(\cdot, t)$ has N simple real roots $z_1(t) < \cdots < z_N(t)$. They obey*

$$t > 0: \quad x_i < z_i(t) < y_i \ (i < N), \qquad x_N < z_N(t), \tag{2}$$
$$t < 0: \quad z_1(t) < x_1, \qquad y_i < z_{i+1}(t) < x_{i+1} \ (i < N). \tag{3}$$

Every $\dot z_i$ is positive, $\sum_i \dot z_i = V := \sum_i v_i$, and

$$\ddot z_i = 2\sum_{j\neq i} \frac{\dot z_i \dot z_j}{z_i - z_j}. \tag{4}$$

Thus (1) is a complete collision-free goldfish solution with $z_i(0) = x_i$ and $\dot z_i(0) = v_i$.

Proof. Off the poles x_i, partial fractions give

$$R(z) := \frac{Q(z)}{P(z)} = \sum_{i=1}^{N} \frac{v_i}{z - x_i}, \qquad R'(z) = -\sum_{i=1}^{N} \frac{v_i}{(z - x_i)^2} < 0. \tag{5}$$

Hence $R = 0$ has exactly one root y_i in each interval (x_i, x_{i+1}). When $t \neq 0$, $p(z,t) = 0$ is equivalent to $R(z) = 1/t$. Strict decrease and the one-sided pole limits give exactly the intervals in (2)–(3), including the sign-selected exterior interval. All roots are simple. At $t = 0$ they are the simple roots of P, so this construction covers all real time.

For $t \neq 0$, differentiation of $R(z_i(t)) = 1/t$ gives

$$\dot z_i = -\frac{1}{t^2 R'(z_i)} > 0. \tag{6}$$

At zero, implicit differentiation instead gives $\dot z_i(0) = Q(x_i)/P'(x_i) = v_i$. The coefficient of z^{N-1} in p gives the exact law

$$\sum_i z_i(t) = \sum_i x_i + Vt, \tag{7}$$

and therefore $\sum_i \dot z_i = V$. Differentiating $p(z,t) = \prod_i (z - z_i(t))$ in t shows that the time-independent Q is

$$Q(z) = \sum_i \dot z_i(t) \prod_{j\neq i}(z - z_j(t)). \tag{8}$$

At a root z_i, elementary logarithmic differentiation gives

$$\frac{p_{zz}}{p_z} = 2\sum_{j\neq i} \frac{1}{z_i - z_j}, \tag{9}$$

$$\frac{Q'}{p_z} = \sum_{j\neq i} \frac{\dot{z}_i + \dot{z}_j}{z_i - z_j}. \tag{10}$$

Twice differentiating $p(z_i(t), t) = 0$ and inserting (9)–(10) yields (4). Finally, $p(z, t+s) = p(z,t) - sQ(z)$ proves the exact flow group law. $\square$

Source lineage

References

[1] F. Calogero, “The ‘neatest’ many-body problem amenable to exact treatments (a ‘goldfish’?),” *Physica D* 152–153 (2001), 78–84. DOI: 10.1016/S0167-2789(01)00160-9.

[2] F. Calogero, “Motion of poles and zeros of special solutions of nonlinear and linear partial differential equations, and related ‘solvable’ many-body problems,” *Nuovo Cimento B* 43 (1978), 177–241. Handle: 11573/11324.

[3] M. C. Nucci, “Calogero’s ‘goldfish’ is indeed a school of free particles,” *Journal of Physics A* 37 (2004), 11391–11400. DOI: 10.1088/0305-4470/37/47/008.